

A Governed Zero Trust AI Gateway

Private AI. Strong identity. Scoped knowledge. Auditable access. Clean offboarding.

The Problem

Most AI deployments trade governance for speed. Shared API keys, no per-user identity, no audit trail, no offboarding procedure. When an engagement ends, access persists until someone revokes it.

Self-hosted AI solves the sovereignty problem but introduces new gaps: inference without identity enforcement, ungoverned knowledge access, and no session controls. Capability without governance is not a security posture.

What Was Built

- Tailscale WireGuard mesh: peer-to-peer where possible, encrypted DERP fallback
- Nginx Proxy Manager with auth_request: Zero Trust enforcement point
- Authelia MFA: WebAuthn, Touch ID, TOTP before any AI access
- **OpenWebUI**: role-scoped RAG KB access, user-scoped conversation history
- Ollama on MacBook Air M4: local inference, no third-party data exposure
- pf firewall: Ollama API restricted to localhost and Umbrel only, enforced at boot via LaunchDaemon
- AdGuard Home with Quad9 DoH: DNS filtering and encrypted upstream resolution
- Offboarding: 4-step revocation across Tailscale, Authelia, and OpenWebUI
- **AI security lab**: prompt injection, RAG poisoning, and AIBOM recovery tests mapped to MITRE ATLAS and NIST AI RMF

Result: Private AI access with device trust, MFA, scoped knowledge, audibility, and clean 4-step offboarding.

Zero Trust Controls at a Glance

Identity	Network	Data	Operations
• Authelia MFA • WebAuthn / Touch ID / TOTP • Admin-mediated provisioning • Dual-step enrollment	• Tailscale WireGuard mesh • No public exposure • Device authorization required • DERP relay fallback	• Role-scoped RAG KBs • User-scoped log history • AIBOM-documented assets • Instruction-augmented RAG	• 4-step offboarding • pf firewall on inference API • AdGuard DNS filtering • LaunchDaemon persistence

Identity Provisioning Model (No Self-Service)

No user can self-provision. Every account requires direct administrator action across two systems before access is granted.

- Admin manually creates user in Authelia YAML file
- Hashed password delivered out of band to the user
- Admin SSHs into Authelia Docker container to retrieve enrollment OTP
- OTP delivered to user: completes MFA enrollment via WebAuthn, Touch ID, or TOTP
- Account becomes operational only after all steps complete
- Ollama inference runs locally on MacBook only. Instance is fully offline-capable; Umbrel version tied to MacBook instance.

Known Gaps (Documented)

- Certificate lifecycle: manual renewal required before Aug 25 2026.
- Post-quantum: Curve25519 handshake not quantum-safe. PSK support tracked on Tailscale roadmap.
- No dynamic PDP: trust evaluation is administrator-mediated at provisioning. Dynamic per-request evaluation is a future phase item.